

TRIAGE-LINK — Master Technical Architecture

Project: TRIAGE-LINK (repo: `Emergency-Medical-Support-System`) **Type:** Offline-first Progressive Web App for casualty care & transport documentation, with a conflict-aware sync backend and a provincial-EHR integration layer (Ontario Health).
Status: ⚠️ **Prototype — not a medical device and not for clinical use.** Several production concerns (auth, encryption at rest, EHR onboarding) are designed-for but not production-hardened; this is called out per section.

This is the **master** architecture document. It consolidates the system into the sections below. Two companion documents go deeper on specific areas and remain authoritative there:

- `docs/ARCHITECTURE.md` — narrative "as-built vs target" design, body-model internals, diagrams.
- `docs/DEPLOYMENT.md` — environment-by-environment deployment detail.

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1. Overview

TRIAGE-LINK is a digital replacement for the paper triage tag used by emergency responders at accident scenes and mass-casualty incidents (MCIs). A single responder — often with no network connectivity — captures a casualty's identity, injuries (marked

on a 2-D body chart), vital signs, and treatments at the point of care, then hands the record to a receiving hospital as a standards-compliant **HL7 FHIR R4** bundle.

The system has three cooperating tiers:

Tier	What it is	Connectivity
Field client	The PWA — capture UI, on-device store, op-log	Works fully offline
Sync service	Fastify + PostgreSQL backend: conflict-aware record reconciliation + EHR gateway	Online
Provincial EHR	Ontario Health's province-wide EHR via the ONE Access Gateway	Online, gated

The defining constraint is **offline-first**: the scene may have no signal, so the device is authoritative locally and reconciles later. The defining interoperability choice is **HL7 FHIR R4**, so hospital EHRs and the provincial registry understand the record without a bespoke format. The interface is **multilingual** (English / French / Arabic / Persian, including right-to-left), so a responder captures and hands over in their own language.

Primary actors

Actor	Role
Field responder (paramedic, medic, first-aider)	Captures the casualty record on a phone/tablet, usually offline
Receiving facility (hospital EHR)	Imports the FHIR bundle at handover
Provincial EHR (Ontario Health)	Confirms patient identity (PCR), supplies clinical context, receives contributions
System operator	Runs the sync service; reviews the EHR access audit trail

2. Requirements

2.1 Functional requirements

ID	Requirement	Status
F-1	Capture patient identity ("tombstone": name, DOB, sex, MRN, blood type, NOK)	✓
F-2	Record incident details (time, mechanism, location, START triage category, age band)	✓
F-3	Mark injuries on an interactive anterior/posterior body chart with auto-detected body region	✓
F-4	Per-injury severity and notes; injury-type palette	✓
F-5	Record timestamped vitals (HR, BP, RR, SpO ₂ , GCS, pain)	✓
F-6	Log treatments (type, detail, place, provider)	✓
F-7	Age-adjusted burn TBSA (Lund-Browder) computed live from burn injuries	✓
F-8	Persist all data on-device; auto-save with no explicit "save" step	✓
F-9	Manage multiple casualties; triage board; printable casualty summary	✓
F-10	Export a FHIR R4 bundle for hospital handover	✓
F-11	Multi-device sync with deterministic, conflict-aware merge (no lost writes)	✓ (backend)
F-12	Verify casualty identity against the provincial client registry (PCR <code>Patient/\$match</code>)	✓
F-13	Pull clinical context (meds/allergies/labs) for a resolved patient	✓
F-14	Contribute a casualty handover to the EHR (write) where entitled	✓
F-15	Durable, queryable audit trail of every EHR access (ATNA AuditEvent)	✓
F-16	Multilingual UI (English / French / Arabic / Persian) with persistent choice and right-to-left layout (Arabic & Persian); selectable via in-app toggle or a <code>?lang=</code> URL switch	✓
F-17	Live time-since-injury clock (<code>T+</code> elapsed) on the acuity glance, triage board, and casualty card	✓
F-18	Handover sign-off (receiving clinician + facility, auto-timestamped) reflected on the saved list, triage board, and casualty card	✓
F-19	Triage board search (name / ID / mechanism / location) and on-scene vs handed-over filtering	✓
F-20	Guided, voiced onboarding tour; offline EHR Test Lab (dev/QA) exercising the real gateway in-browser	✓

2.2 Non-functional requirements

Attribute	Target
Offline capability	100% of capture functions usable with zero connectivity
Portability	One codebase runs on any modern browser; installs to the home screen
Sync latency	Record reaches the central tier within seconds of connectivity returning
Data integrity	No silent data loss; all merge conflicts resolved deterministically and audited
MCI burst	Many concurrent responders without capture-side degradation
Handover time	Scan/transfer-to-EHR in seconds, not minutes
Availability	≥ 99.9% for central services; handover path prioritised
Auditability	Every PHI/EHR access is logged immutably — who, what, when
Fail-closed safety	The system never serves fabricated data on misconfiguration

2.3 Constraints & assumptions

- **No PHI leaves the device** unless the user explicitly exports a bundle or sends to the EHR.
- The PWA **cannot** hold ONE ID secrets or present a client certificate — all credentialed EHR traffic goes through the backend.
- Connecting to Ontario's **production** EHR requires Health Information Custodian onboarding, ONE ID credentials, mutual TLS, and conformance testing — out of scope for the prototype, but the interfaces are built to accept them as configuration.

3. Use cases

UC-1 — Capture a casualty offline

Actor: Field responder. **Precondition:** No connectivity.

1. Responder opens the installed PWA (loads from the service-worker cache).
2. Enters tombstone + incident; taps the body chart to mark injuries; records vitals and treatments.
3. Each edit re-renders immediately and auto-saves to IndexedDB after a 400 ms debounce. **Outcome:** A durable `CasualtyRecord` on-device; no data lost if the app closes.

UC-2 — Hand over to a hospital via FHIR export

Actor: Field responder → receiving facility.

1. Responder taps **Export FHIR ↓**.

2. `toFhirBundle(record)` produces a FHIR R4 `collection` bundle; the browser downloads it. **Outcome:** A standards-compliant bundle the hospital EHR can ingest. Pure client-side; nothing transmitted.

UC-3 — Reconcile the same casualty across devices

Actor: Two responders editing one casualty.

1. Each device journals edits as Lamport-ordered **ops** in an append-only log.
2. On connectivity, `POST /sync` ingests ops idempotently and folds them with the deterministic resolver.
3. Edits to different fields merge cleanly; same-target edits pick a Lamport winner and the loser is **retained** and reported as a conflict. **Outcome:** Byte-identical converged state on every replica; full audit trail; no lost writes.

UC-4 — Verify identity against the provincial registry (PCR)

Actor: Field responder (via backend).

1. Responder enters a health-card number (or name + DOB) and taps **⊕ Verify identity against PCR**.
2. The PWA calls `POST /ehr/patient/$match`; the backend builds a FHIR `Parameters`, calls PCR via the ONE Access Gateway, and returns graded matches.
3. Responder applies a matched identity onto the record (promotes the registry id/health-card number). **Outcome:** Confirmed patient identity; every query audited.

UC-5 — Pull clinical context for a known patient

Actor: Field responder (via backend).

1. After a resolved match, responder taps **Load clinical context**.
2. `GET /ehr/patient/:id/context` fetches meds/allergies/labs (DHDR/OLIS/Patient Summary), merging entitled repositories into one collection bundle (repositories the caller isn't entitled to are skipped). **Outcome:** Allergies/medications visible at the point of care.

UC-6 — Contribute a handover to the EHR

Actor: Field responder (via backend), where entitled.

1. Responder taps **Send to EHR ↑**.
2. `POST /ehr/handover` maps the record to an Ontario-profiled **transaction** bundle (OHIP identifier) and POSTs it. Writes are **never retried** (no duplicate contribution). **Outcome:** The handover is contributed (or a clear failure is surfaced); audited as a create.

UC-7 — Review the EHR access audit trail

Actor: System operator.

1. Operator calls `GET /ehr/audit` (optionally `?patient=`).

2. The service returns persisted ATNA AuditEvents (action/outcome/agent/patient/query + full FHIR resource), newest first. **Outcome:** Accountable, durable record of every provincial-EHR access.

UC-8 — Document in the responder's language (multilingual / RTL)

Actor: Field responder. **Precondition:** None (works fully offline).

1. The interface loads in the responder's language — chosen from the header  toggle (cycles English → French → Arabic → Persian) or forced by a `?lang=ar` / `?lang=fa` URL switch (handy for kiosks/QR links). The choice persists to `localStorage`, so it sticks until switched.
2. For Arabic and Persian, the document direction flips to right-to-left; the whole clinical UI, the guided tour voice-over, the anatomical region names, and the printable casualty card render in-language. Patient data (names, IDs, treatment/injury keys) stays language-neutral so the record and its FHIR export are unaffected. **Outcome:** A field responder captures and hands over entirely in their own language without changing the underlying record.

UC-9 — Sign off and share a handover

Actor: Field responder → receiving facility.

1. At the receiving end, the responder enters the receiving clinician and facility and taps ✓ **Mark handed over**; the time is stamped automatically. The casualty is now flagged "handed over" on the saved list, on the triage board (filterable, with who/where/when on the card), and on the casualty card.
2. Signing closes the FHIR **Encounter** (`period.end` , an attender participant, the receiving `serviceProvider`) and emits a **Provenance** documenting the transfer. The responder can ↪ **Share handover** to download a focused FHIR slice — Patient + Encounter + Provenance only. **Outcome:** An auditable, standards-based record of *who took over care, when, and where* — exportable on its own or as part of the full bundle.

4. Technology used

4.1 Field client (PWA)

Concern	Choice	Version	Rationale
UI framework	React + TypeScript	18.3	Component model for capture UI; static typing across the domain
Build / dev	Vite	8.1	Fast dev server; simple static production build
Offline storage	IndexedDB via Dexie	4.0	Durable, async, typed on-device store; no backend needed
Offline shell	vite-plugin-pwa (Workbox)	1.3	Service worker precaches the app for offline use
Localization	In-house React-context i18n (EN/FR/AR/FA, RTL)	—	No dependency; offline-first; flat dictionaries with English fallback; <code>?lang=</code> URL switch
Interop	HL7 FHIR R4	—	De-facto hospital/registry exchange standard

4.2 Backend (sync service + EHR gateway)

Concern	Choice	Version	Rationale
HTTP server	Fastify	5.8	Lightweight, typed, fast
System of record	PostgreSQL via pg	8.12	Durable relational store for ops/snapshots/audit
EHR auth	ONE ID (OAuth 2.0 client-credentials / OIDC)	—	Ontario Health's identity & access service
EHR transport	ONE Access Gateway + mutual TLS	—	The front door to Ontario's provincial repositories
Identifiers	OHIP health card (FHIR NamingSystem)	—	Provincial patient identity

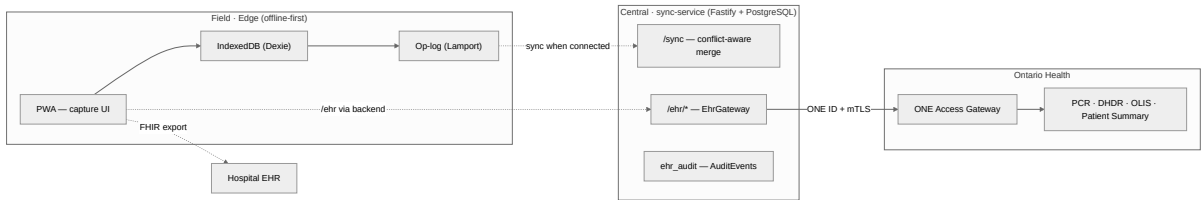
4.3 Shared & tooling

Concern	Choice	Version
Language	TypeScript	5.6
Monorepo	npm workspaces	—
Tests	Vitest (+ pg-mem for in-memory Postgres)	4.1 / 3.0
CI/CD	GitHub Actions (CI + Pages deploy)	—

There are intentionally **no heavyweight runtime dependencies** in the core (no FHIR library): the domain model, FHIR mapping, op-log, and EHR port are hand-written framework-free TypeScript.

5. System architecture

5.1 Three-tier topology



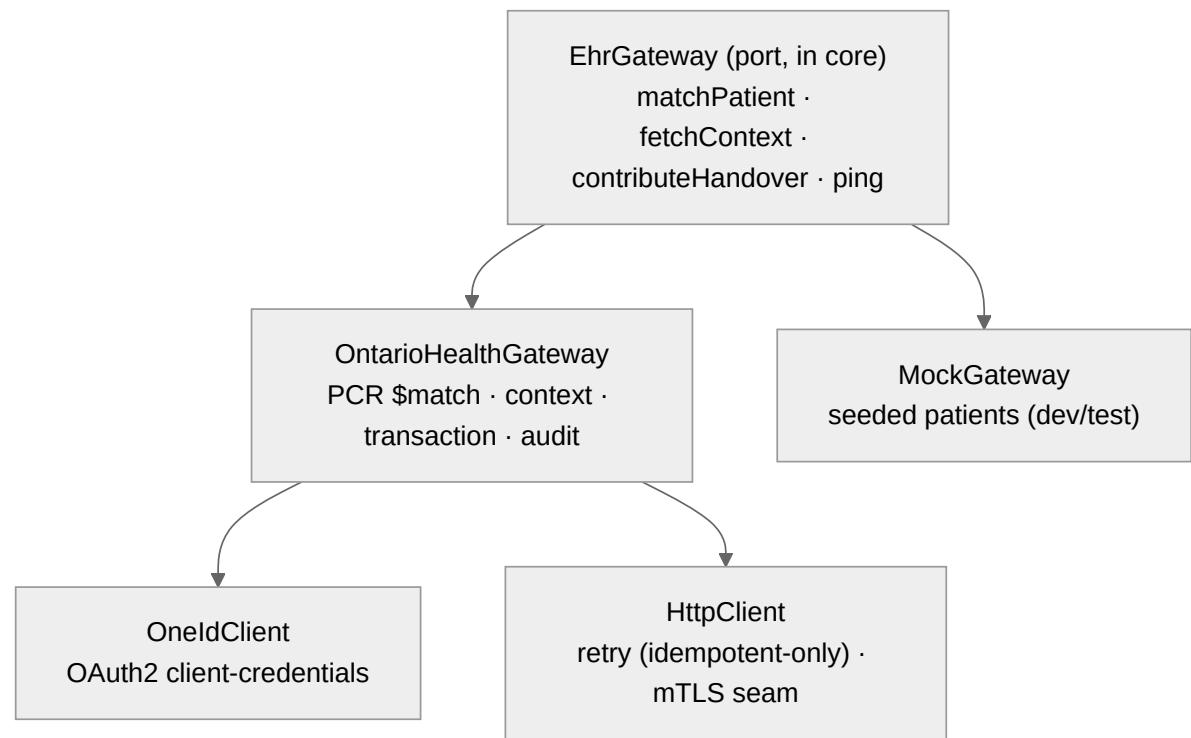
5.2 Monorepo packages & dependency direction

Dependencies point **inward**: UI → core; backend → core; nothing depends on the UI.

@trriage-link/core	framework-free: domain, fhir, sync (op-log), ehr (port + C
@trriage-link/ehr-gateway	server-side EHR adapters: ONE ID, OntarioHealthGateway, Mo
@trriage-link/sync-service	Fastify + PostgreSQL: /sync, /ehr/*, audit store
src/ (root)	the PWA app – consumes @trriage-link/core

5.3 EHR integration as a port/adaptor

The app depends only on the provider-agnostic `EhrGateway` interface (in core). Ontario Health is one concrete adapter; `MockGateway` backs dev/tests. Adding a province = implementing the same interface — nothing upstream changes.



6. Data model

6.1 The casualty record (single source of truth)

The entire clinical record for one patient is one `CasualtyRecord`

(`packages/core/src/domain/types.ts`):

```
CasualtyRecord
├─ id                stable case ID (also seeded as the MRN)
├─ tombstone         Tombstone   name, dob, sex, mrn, bloodType, address, nextOfKin, nextC
├─ incident          Incident    injuryTime, mechanism, location, triage, ageBand
├─ injuries          Injury[]    view, x/y, region, type, severity, notes
├─ vitals            VitalSign[] takenAt + hr/bp/rr/spo2/gcs/pain
├─ treatments        Treatment[] performedAt, type, detail, place, provider
├─ handover          Handover?   at, clinician, facility (null until handover)
├─ createdAt         number
└─ updatedAt         number
```

- **One record = one unit of sync** (the op-log reconciles per record).
- **"Tombstone" = stable identity layer**, separated from the episode-specific `incident`, mirroring how hospital systems separate a Patient from an Encounter.
- **Triage** uses START — immediate (red) / delayed (yellow) / minor (green) / deceased (black).

6.2 Op-log (sync) model

Each edit becomes an immutable **op** — `scalar` (a field path), `item-put`, or `item-remove` — carrying a **Lamport clock**, a `clientId`, and a unique id (idempotency key). `resolve(recordId, ops)` folds ops in canonical order `(lamport, clientId, id)`, so any two replicas with the same ops compute byte-identical state. Losing ops are retained and reported as `ConflictReport`s.

6.3 FHIR mapping (handover)

`packages/core/src/fhir/mapping.ts` maps the record to a FHIR R4 bundle:

Domain concept	FHIR resource	Notes
Tombstone	Patient	name, gender, birthDate, NOK contact; identifier <code>urn:trriage-link:case</code>
Incident/episode	Encounter	<code>class=EMER</code> ; status <code>in-progress</code> → <code>finished</code> ; mechanism → <code>reasonCode</code> . A signed handover adds <code>period.end</code> , an ATND attender participant, and the receiving <code>serviceProvider</code>
Handover	Provenance	emitted only once signed — <code>recorded</code> time, a <code>TRANSFER</code> activity, a custodian agent (clinician <code>onBehalfOf</code> facility) targeting the Encounter
Injury	Condition	category <code>injury</code> ; <code>bodySite</code> = region+view; severity; notes
Vital	Observation	category <code>vital-signs</code> ; LOINC-coded ; value + unit
Treatment (non-drug)	Procedure	status <code>completed</code> ; performer; detail + place
Treatment (drug)	MedicationAdministration	when the intervention matches <code>/medication/i</code>

`toFhirBundle` emits the full `collection`; `toHandoverBundle` returns the focused slice (Patient + Encounter + Provenance) behind the ↪ **Share handover** action.

6.4 Ontario-profiled FHIR (registry)

`packages/core/src/ehr/ontario.ts`:

- `ONTARIO_SYSTEMS` — OHIP health-card identifier systems.
- `buildPatientMatchParameters` — FHIR `Parameters` for PCR `Patient/$match`.
- `parsePatientMatchBundle` — reads the authoritative FHIR `match-grade` extension (falling back to the numeric `search.score` only when absent); resolves identity only on a single `certain` candidate.
- `toOntarioContributionBundle` — a FHIR **transaction** bundle with the OHIP patient identifier for contribution.
- `buildAccessAuditEvent` — an ATNA `AuditEvent` per access.

6.5 Persistence schema (PostgreSQL)

Table	Purpose
ops	Append-only operation log (idempotent ingest)
snapshots	Resolved per-record state
audit	Sync op/conflict events (op-ingested, conflict-resolved)
ehr_audit	EHR-access AuditEvents — full FHIR resource + queryable columns (action/outcome/agent/patient/query)

7. Implementation

7.1 Field client (src/)

- **App.tsx** — capture UI; immutable state mutators; 400 ms debounced auto-save via `recordRepo`; the handover sign-off panel and the FHIR export / share-slice downloads.
- **i18n.tsx** — in-house React-context i18n: flat EN/FR/AR/FA dictionaries with English fallback, `t(key, params)`, `regionLabel()` for localised anatomy, a `?lang=` URL switch that wins over the persisted choice, and `document.dir = rtl` for Arabic. `useNow.ts` drives the live elapsed clock.
- **components/BodyChart.tsx** — anterior/posterior injury-marking SVG; the same polygons are drawn and hit-tested, so the figure and tappable regions never drift.
- **components/PcrVerify.tsx** — PCR identity verification UI in the Tombstone panel: builds the query from the tombstone, lists graded matches, applies a chosen identity, loads context.
- **components/TriageBoard.tsx**, **CasualtySummary.tsx**, **Elapsed.tsx** — multi-casualty board (searchable, on-scene / handed-over filter, handover detail per card); printable AT-MIST card; the live time-since-injury clock.
- **db/ (Dexie)** — `records` + append-only `ops` + `meta` (clientId + Lamport); `repository.ts` journals ops in the same transaction as the record write.
- **ehr/client.ts** — browser client for the backend EHR routes (`VITE_EHR_BASE_URL`), with an offline/unreachable error path; the PWA never calls Ontario directly.

7.2 Core (packages/core/)

Framework-free, built to `dist/` (ESM + `.d.ts`) and consumed by both the PWA and the backend:

- **domain/** — `CasualtyRecord` and sub-types; injury catalog; body model + region hit-testing; burn TBSA; id generation.
- **fhir/** — minimal R4 types; `toFhirBundle`.
- **sync/** — `diffToOps`, `mergeOps`, `resolve` (deterministic Lamport-ordered fold).
- **ehr/** — `EhrGateway` port, `PatientIdentity` / `MatchResult` / `ContributionResult` types, typed `EhrError`, `identityFromTombstone`, and the Ontario mappings (§6.4).

7.3 EHR gateway (`packages/ehr-gateway/`)

- `OntarioHealthGateway` — implements `EhrGateway` against the ONE Access Gateway:
 - `matchPatient` → PCR `Patient/$match` (POST, opted-in as idempotent → retryable).
 - `fetchContext` → searches configured repositories `{type}?patient={id}`, skips 403/404 (not entitled), merges into one collection bundle.
 - `contributeHandover` → POSTs an Ontario transaction bundle, **no retry** (no duplicate write), audited as action `C`.
 - `ping` → `CapabilityStatement`.
 - Emits an ATNA `AuditEvent` on every access; transparent token refresh on 401.
- `OneIdClient` — OAuth 2.0 client-credentials token client; caches the token and **coalesces concurrent refreshes** onto a single in-flight fetch.
- `HttpClient` — typed `EhrError`s; exponential-backoff retries **gated to idempotent requests only** (GET, or explicit opt-in) so writes are never replayed; an mTLS `dispatcher` seam.
- `MockGateway` — in-memory provider with seeded patients/context; records contributions.

7.4 Sync service (`packages/sync-service/`)

- `app.ts` / `buildApp` — wires routes; EHR routes mount only when a gateway is injected.
- `ops-store.ts` — append-only ops/snapshots/audit over a `Queryable` seam (prod `pg`, tests `pg-mem`).
- `ehr-routes.ts` — `GET /ehr/health`, `POST /ehr/patient/$match` (body validated), `POST /ehr/handover`, `GET /ehr/patient/:id/context`, `GET /ehr/audit`; maps `EhrError` codes to HTTP status.
- `ehr-audit-store.ts` — `ehr_audit` table + `EhrAuditStore` (record/list, patient filter).
- `server.ts` — production entrypoint; **fail-closed** EHR provider selection (see §9.3).

7.5 HTTP API summary

Method & path	Purpose
<code>GET /health</code>	Liveness
<code>POST /sync</code>	Ingest ops idempotently; return resolved state + ops
<code>GET /sync/:recordId</code>	Snapshot + op-log + audit for a record
<code>GET /ehr/health</code>	EHR provider liveness/auth probe
<code>POST /ehr/patient/\$match</code>	Resolve identity against PCR
<code>GET /ehr/patient/:id/context</code>	Pull meds/allergies/labs
<code>POST /ehr/handover</code>	Contribute a handover (write)
<code>GET /ehr/audit</code>	Read the EHR access audit trail

8. Test cases

All suites run under **Vitest**; the backend uses **pg-mem** (in-memory PostgreSQL) so DB tests run with identical SQL and no external service. **155 tests pass** across the workspaces (core 68, field client 35, sync 33, EHR gateway 19).

8.1 Core (`packages/core/test/` , 68 tests)

Suite	Covers
<code>fhir-mapping.test.ts</code>	Record → FHIR bundle: Patient/Encounter/Condition/Observation (LOINC)/Procedure/MedicationAdministration; handover closes the Encounter (<code>period.end</code> , <code>attender</code> , <code>serviceProvider</code>) and emits a Provenance ; <code>toHandoverBundle</code> slice
<code>elapsed.test.ts</code>	Time-since-injury maths — elapsed parsing (null on empty/future) and compact <code>T+</code> formatting with localisable units
<code>sync.test.ts</code>	<code>diffToOps</code> / <code>mergeOps</code> / <code>resolve</code> — deterministic ordering, conflict retention, convergence
<code>regions.test.ts</code>	Body-region hit-testing + anatomical sidedness
<code>injuries.test.ts</code> , <code>types.test.ts</code> , <code>id.test.ts</code>	Catalog, factories, id generation
<code>ehr.test.ts</code>	<code>identityFromTombstone</code> ; <code>\$match</code> Parameters; match-grade extension is authoritative ; <code>certainly-not</code> with a high score does not resolve; AuditEvent builder

8.1b Field client (`test/` , 35 tests, jsdom)

Suite	Covers
<code>app.test.tsx</code>	Core capture flows; vitals + GCS; AT-MIST card; DOB → age; calendar popup; triage from the header; language toggle persists ; <code>?lang=ar</code> and <code>?lang=fa</code> URL switches flip to Arabic / Persian + RTL ; handover sign-off → confirmation → undo
<code>board.test.tsx</code>	Triage-board search (<code>matchesQuery</code>) and the on-scene / handed-over filter; handover detail on cards
<code>i18n.test.ts</code>	EN/FR/AR/FA dictionary parity (no missing keys / English leaks); the four-language cycle; RTL flags; region-name localisation incl. Arabic & Persian
<code>ehr-console.test.tsx</code> , <code>db.integration.test.ts</code>	EHR Test Lab console; IndexedDB repository round-trips

8.2 EHR gateway (`packages/ehr-gateway/test/` , ~19 tests)

Suite	Covers
<code>http.test.ts</code>	Idempotent GET retries on 503; non-idempotent POST is not retried ; explicit opt-in retries; non-retryable 403
<code>one-id.test.ts</code> (within suite)	Client-credentials token + caching; concurrent refreshes coalesce to one fetch
<code>ontario-health-gateway.test.ts</code>	Authenticated <code>\$match</code> + parse; underspecified query rejected pre-flight; 401 → refresh + retry once ; context merge skipping 403; contribution is a transaction bundle, not retried ; failure audited
<code>mock-gateway.test.ts</code>	Certain/probable/no-match; seeded context; contribution recorded

8.3 Sync service (`packages/sync-service/test/` , 33 tests)

Suite	Covers
<code>sync.integration.test.ts</code>	Two offline clients converge through <code>/sync</code> ; idempotent ingest; conflict audit
<code>ehr-routes.integration.test.ts</code>	<code>/ehr/health</code> ; <code>\$match</code> resolve + 400 on non-object body ; context bundle; handover + 400 without id ; EHR routes unmounted when no gateway; end-to-end audit (match → persisted → <code>GET /ehr/audit</code>)
<code>ehr-audit.test.ts</code>	Persist/read AuditEvent with extracted columns; patient filter + newest-first; limit cap

8.4 Test strategy

- **Deterministic by construction** — injected clocks/fetch (no real time/network/DB).
- **Same code path in mock and prod** — `MockGateway` round-trips through the production parser.
- **Adversarial paths covered** — token expiry, 401/403/503, malformed bundles, oversized limits, no-double-write on failed writes.

9. Deployment

9.1 Field client (static PWA)

`npm run build` type-checks and emits a self-contained static `/dist` bundle; the service worker makes it work offline after first load. Deploy to any static host/CDN:

- **GitHub Pages** — `.github/workflows/deploy-pages.yml` builds with `GITHUB_PAGES=true` (sets the `/Emergency-Medical-Support-System/` base path) and publishes `dist/` on push to `main` .
- **Netlify / S3 / nginx** — drop the `dist/` folder; base path stays `/` .

- `VITE_EHR_BASE_URL` (build-time) points the PWA at the backend EHR routes (empty = same origin).

Web security headers. The production build injects a strict **Content-Security-Policy** as a `<meta>` tag (`default-src 'self'`, `object-src 'none'`, `script-src 'self'`, `img-src 'self' data: blob:`, `style-src 'self' 'unsafe-inline'` for React inline styles; `connect-src` gains `VITE_EHR_BASE_URL`'s origin when set) — so it applies even on GitHub Pages, which can't set response headers. The remaining headers a meta tag can't set (HSTS, `X-Frame-Options: DENY`, `Referrer-Policy`, `Permissions-Policy` locking the camera to `self`, COOP/CORP) ship in `public/_headers` for Netlify/Cloudflare Pages; nginx/Apache operators should mirror them. On GitHub Pages only the meta CSP applies (no custom headers, so no `X-Frame-Options` there). The dev-only **EHR Test Lab** is gated on `import.meta.env.DEV` and is **tree-shaken out of the production bundle** (no URL flag reaches it).

9.2 Sync service (container)

- Run `node --experimental-strip-types src/server.ts` (or compile first).
- `DATABASE_URL` — PostgreSQL connection string; `migrate()` + `migrateEhrAudit()` create tables on boot.
- `PORT` — listen port (default 8080).
- Stateless and horizontally scalable; PostgreSQL is the managed stateful tier.

Transport/access hardening (all optional; the API ships open for dev/tests, so a production deploy MUST set the first two):

Variable	Effect
<code>SYNC_API_TOKEN</code>	When set, <code>/sync</code> and <code>/ehr/*</code> require <code>Authorization: Bearer <token></code> (constant-time check). <code>/health</code> stays open.
<code>CORS_ORIGINS</code>	Comma-separated browser-origin allowlist (the PWA's origin). Unset → cross-origin requests are blocked.
<code>RATE_LIMIT_MAX</code>	Per-IP requests/minute (default 300).
<code>TRUST_PROXY</code>	<code>true</code> behind a reverse proxy / load balancer (honours <code>X-Forwarded-*</code>).
<code>ENABLE_DOCS</code>	<code>true</code> to serve Swagger UI at <code>/docs</code> (off by default in production).

Always-on: hardened response headers (`@fastify/helmet` — `nosniff`, `X-Frame-Options: DENY`, HSTS, no-referrer), a 10 MB request-body cap, and JSON-schema validation of the `/sync` op batch (bounded size; unknown fields stripped).

9.3 EHR provider configuration (fail-closed)

`server.ts` selects the EHR adapter from the environment and **fails closed** so fabricated mock data is never served by accident:

Condition	Result
All Ontario vars present	Real <code>OntarioHealthGateway</code>
Some but not all present	Throws at startup (catches a misconfigured prod deploy)
None present + <code>EHR_ALLOW MOCK=true</code>	<code>MockGateway</code> (explicit dev opt-in)
None present	EHR routes are not mounted

Variable	Meaning
<code>ONE_ID_TOKEN_URL</code>	ONE ID OIDC token endpoint
<code>ONE_ID_CLIENT_ID</code> / <code>ONE_ID_CLIENT_SECRET</code>	client-credentials
<code>ONE_ID_SCOPE</code>	granted scopes (e.g. <code>pcr/Patient.read</code>)
<code>OH_FHIR_BASE_URL</code>	ONE Access Gateway FHIR base
<code>OH_AGENT_ID</code>	requesting clinician id for the audit trail
<code>EHR_ALLOW MOCK</code>	<code>true</code> to serve the in-memory mock in dev

Mutual TLS: construct an undici `Agent` with the client certificate and pass it as the `dispatcher` to `OneIdClient` / `OntarioHealthGateway` (a documented seam in `server.ts`).

9.4 CI/CD

`.github/workflows/ci.yml` (on PRs to `main`): `npm ci` → root typecheck → typecheck each of `core` / `ehr-gateway` / `sync-service` → `npm test` → `npm run build`.

9.5 Local development

```
npm install      # installs workspaces; builds @trriage-link/core (prepare)
npm run dev      # Vite dev server at http://localhost:5173
npm test        # all workspace tests (Vitest)
npm run build    # type-check + production build to /dist

# Backend (mock EHR, no Ontario credentials needed):
EHR_ALLOW MOCK=true DATABASE_URL=postgres://... \
  npm start --workspace @trriage-link/sync-service
```

9.6 Target production topology

Stateless services on managed Kubernetes behind a WAF/CDN and load balancer; a message queue absorbs MCI write bursts and reconnection storms; managed PostgreSQL (primary + replica), a FHIR server, and a WORM audit store form the data tier. The hospital reaches the FHIR endpoint via SMART-on-FHIR; the provincial EHR via the ONE Access Gateway with mutual TLS. (See `docs/ARCHITECTURE.md` §13 for the full diagram. Not yet implemented.)

10. Security & privacy

Status: prototype (partially hardened). Data is local to the device; wound photos and exported backups can be encrypted at rest (AES-256-GCM/PBKDF2), and the sync backend is auth/rate-limit/CORS-gated. There is still no end-user auth on the PWA, and record text fields are not yet encrypted. A production deployment applies defense-in-depth:

Domain	Requirement	Priority
Encryption	TLS 1.3 in transit; AES-256 at rest (device, DB, backups); mTLS to the EHR gateway. <i>Partly built:</i> AES-256-GCM at rest for photos (opt-in vault) and encrypted backups	Must
Device security	Full-disk encryption, screen/biometric lock, MDM, remote wipe	Must
Authentication	OAuth2/OIDC for users; ONE ID for the EHR; SMART-on-FHIR for hospital EHR; MFA for privileged roles	Must
Authorisation	RBAC, least privilege; field/dispatch/clinician/admin scoped separately	Must
Audit logging	Immutable log of every PHI/EHR access — implemented for EHR access (<code>ehr_audit</code> , ATNA AuditEvents)	Must
Data minimisation	Collect only what care requires; no PHI in logs/analytics	Must
Secrets	ONE ID secrets and client certs live only on the backend; never in the PWA, source, or images	Must
Integrity at handover	Signing / provenance on the exported/contributed bundle	Should

Implemented today: opt-in **photo vault** — wound photos encrypted at rest (AES-256-GCM, key from a passphrase via PBKDF2-SHA-256, 210k iterations), with a passphrase lock screen and auto-lock, the key held only in memory; **passphrase-encrypted backups**; sync-service **bearer auth** (constant-time), **rate-limiting**, **CORS allowlist** and `helmet` headers; a client **Content-Security-Policy** and security headers; **CI security scanning** (CodeQL, secret scan, `npm audit`); fail-closed EHR provider selection; idempotent-only retries (no duplicate writes); per-access ATNA audit persisted to PostgreSQL; the PWA never holds EHR credentials.

11. Regulatory & compliance

Area	What it requires	Applies when
PHIPA (Ontario)	Health-information custodian obligations; consent; audit	Ontario patients / EHR access
Ontario Health onboarding	HIC agreements, ONE ID, conformance testing, EHR privacy & security	Connecting to the provincial EHR
HIPAA (US)	Privacy & Security Rules; BAAs; breach notification	US patients/providers
GDPR (EU/UK)	Lawful basis, data-subject rights, DPIA, residency	EU/UK data subjects
SaMD / IEC 62304	Design controls, risk management (ISO 14971), validated SDLC	Software drives clinical decisions
HL7 FHIR conformance	Validate against provincial profiles (PCR IG, Patient Summary)	EHR integration (always)

Decide device classification early. Whether this is a documentation tool or a clinical-decision device changes cost, timeline, and process dramatically.

Before connecting to production Ontario Health

1. Confirm canonical identifier systems/profiles in `ontario.ts` against the onboarded PCR IG version.
2. Supply ONE ID credentials/scopes and the mTLS client certificate.
3. Wire `onAudit` to the audit store (done) and forward to the enterprise SIEM.
4. Pass conformance testing in the Ontario Health sandbox.

12. Roadmap

Because the domain, FHIR, and EHR layers are framework-free, several items add without UI changes:

- **Sync hardening** — end-user auth, encryption of record text fields at rest (photos + backups already encryptable), incremental per-client cursors, cross-device tombstone deletes.
- **EHR breadth** — richer `fetchContext` (typed DHDR/OLIS/Patient Summary resources); SMART-on-FHIR.
- **Profiles** — validate emitted resources against CA Baseline / Ontario PCR profiles.
- **Handover scanning** — NFC/QR handover + master-patient-index reconciliation.
- **Additional provinces** — new `EhrGateway` adapters behind the same port.
- **React Native client** — reuse `@triage-link/core` unchanged.

13. Glossary

Term	Meaning
PWA	Progressive Web App — installable, offline-capable web app
CasualtyRecord	The canonical per-patient model; single source of truth
Tombstone	The stable patient-identity layer (vs the episode-specific incident)
START	Simple Triage And Rapid Treatment — the red/yellow/green/black scheme
TBSA	Total Body Surface Area (burns); Lund-Browder age-adjusted
Op-log	Append-only log of edits, merged with deterministic conflict resolution
Lamport clock	Logical counter giving a total order for deterministic merge
FHIR	HL7 Fast Healthcare Interoperability Resources (R4)
PCR	Provincial Client Registry — Ontario's patient identity registry
DHDR / OLIS	Digital Health Drug Repository / Ontario Laboratories Information System
ONE ID	Ontario Health's identity & access service (OIDC)
ONE Access Gateway	The API gateway fronting Ontario's provincial repositories
OHIP	Ontario Health Insurance Plan — the provincial health-card number
ATNA AuditEvent	The FHIR audit resource recording each access
mTLS	Mutual TLS — client-certificate authentication

*Companion documents: ``docs/ARCHITECTURE.md`` (design narrative & diagrams), ``docs/DEF`